PTZ Control Overview

Video & Streaming - PTZ Control Overview

PTZ (Pan-Tilt-Zoom) Control enables remote operation of motorized cameras, allowing operators to adjust camera position, angle, and zoom level in real-time from the GCXONE interface. This feature is essential for active surveillance, investigation, and dynamic monitoring scenarios.

Key Capabilities

Basic PTZ Operations

- Pan: Horizontal rotation left and right — typically ±360° continuous or limited range
- Tilt: Vertical rotation up and down — typically ±90° or more depending on camera
- Zoom: Optical and/or digital zoom for closer views
- Focus: Automatic or manual focus control on supported cameras
- Iris: Aperture control for light management on supported cameras

Advanced Features

- Preset Positions: Save and recall up to 255 preset camera positions for quick access to critical viewing angles
- Preset Tours: Automated sequences visiting multiple preset positions on a schedule
- Pattern Recording: Record and replay movement patterns for consistent monitoring
- Speed Control: Adjustable movement speed from 1–8 depending on device
- Relative Movement: Move camera by specific increments for fine adjustments
- Absolute Positioning: Move camera to exact coordinates on supported devices
- Rectangle PTZ (Milestone-specific): Draw a rectangle on the screen to optically zoom into that specific area
- Point to Center (Milestone-specific): Click a point on the screen to center the camera on that exact spot

Control Methods

- On-Screen Controls: Click-and-drag or directional buttons in GCXONE interface
- Keyboard Controls: Arrow keys for pan and tilt, +/- for zoom
- Mobile App: Touch-based PTZ controls on mobile devices
- Preset Selection: Quick access to saved preset positions

Supported Devices

PTZ control is supported on PTZ cameras, PTZ domes, speed domes, and cameras with built-in PTZ functionality.

Device requirements:

- Camera must have PTZ hardware capabilities

- Device must support PTZ protocol such as ONVIF or manufacturer SDK
- PTZ functionality must be enabled in device settings
- Appropriate user permissions required

Supported Protocols

- ONVIF PTZ: Standard protocol for PTZ control
- Manufacturer SDKs: Native protocols for Hikvision, Dahua, Axis, and others
- Pelco-D/P: Legacy protocol support
- HTTP/HTTPS: Web-based PTZ control

How It Works

1. Operator initiates PTZ command via GCXONE interface
2. GCXONE translates command to device protocol
3. Command sent to camera via network
4. Camera motor executes movement
5. Camera position and status updated in interface

Use Cases

Active Surveillance

- Monitor moving subjects across large areas
- Follow suspicious activity in real-time
- Respond to alarms by positioning camera on event location

Investigation

- Zoom in on specific details or areas
- Capture multiple angles of incident scene
- Document evidence with preset positions

Site Monitoring

- Routinely scan large areas such as parking lots, warehouses, and perimeters
- Use preset tours for automated monitoring
- Monitor multiple zones with a single PTZ camera

Emergency Response

- Rapidly position camera on alarm location
- Zoom in on emergency situations
- Provide detailed view for security personnel

Best Practices

Preset Management

- Use descriptive names such as "Front Entrance" or "Parking Lot North"

- Assign logical numbers for quick access — 1–10 for critical positions
- Regularly verify preset accuracy and adjust if needed

Movement Control

- Use faster speed for large movements, slower for precision
- Avoid excessive rapid movements to reduce wear on the camera mechanism
- Use relative movement for fine adjustments after reaching a preset

Power Management

- Avoid leaving camera in continuous pan mode unnecessarily
- Configure camera to return to default position after inactivity
- Use presets instead of manual movement when possible to reduce wear

Integration with Alarms

- Configure camera to move to a preset when an alarm triggers
- Create tours that visit alarm locations
- Configure automatic PTZ movement based on events

Limitations and Considerations

Physical Limitations

- Cameras have physical limits to pan and tilt range
- Maximum movement speed varies by camera model
- Digital zoom reduces image quality
- Excessive use may reduce mechanism lifespan

Network Considerations

- Network latency affects real-time control responsiveness
- P2P Local Mode connections provide better PTZ responsiveness

Operational Considerations

- Only one user should control PTZ at a time per camera
- Multiple users changing presets simultaneously may cause conflicts
- Appropriate user permissions required for PTZ